

Shielded Zcash

the Orchard protocol and Halo 2, with the mathematics

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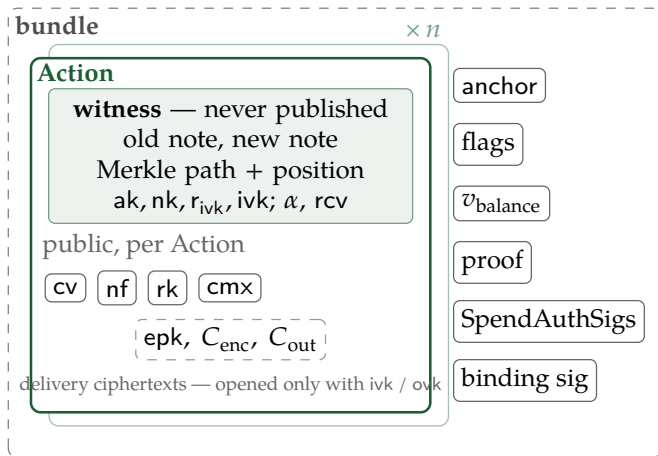
The problem

What a shielded payment must achieve

	soundness: every valid payment	hiding: an observer learns
R1	the record of a created note binds the note	neither address nor value; real = dummy
R2	a consumed note of value $\neq 0$ was committed at the anchor	not which note
R3	no note is consumed twice	no list of consumed notes
R4	$\sum v^{\text{old}} = \sum v^{\text{new}} + v^{\text{balance}}$	only v^{balance}
R5	only the key holder consumes; data it signs cannot change	—

An *observer* reads the chain and holds no key of the notes concerned.

Verification is divided



Per bundle one proof, for one relation per Action, and one binding signature; per Action one spend-authorisation signature; against chain state the anchor and the nullifiers.

Fields, curves, encodings

The Pasta cycle

$$p_{\text{Pallas}} = 2^{254} + 45560315531419706090280762371685220353$$

$$p_{\text{Vesta}} = 2^{254} + 45560315531506369815346746415080538113$$

curve	equation	group order
Pallas	$y^2 = x^3 + 5$ over $\mathbb{F}_{p_{\text{Pallas}}}$	$ \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}) = p_{\text{Vesta}}$
Vesta	$y^2 = x^3 + 5$ over $\mathbb{F}_{p_{\text{Vesta}}}$	$ \mathcal{E}(\mathbb{F}_{p_{\text{Vesta}}}) = p_{\text{Pallas}}$

Both groups have prime order; each curve's scalar field is the other's base field.
Discrete logarithms: about 2^{126} group operations.

Pallas points are the protocol's objects; the proof system commits on Vesta, whose scalars are $\mathbb{F}_{p_{\text{Pallas}}}$.

Encodings and coordinate extraction

$$\text{LE}_k(m) := b_0 b_1 \cdots b_{k-1} \text{ with } m = \sum_i b_i 2^i$$

The map l2LEOSP_k sends the same integer to $\lceil k/8 \rceil$ little-endian bytes.

$$O^* := \text{LE}_{256}(0), \quad (x, y)^* := \text{LE}_{256}(x + 2^{255}(y \bmod 2))$$

$$\text{Extract}_{\mathbb{P}}(O) := 0, \quad \text{Extract}_{\mathbb{P}}((x, y)) := x$$

Lemma. *The map $\text{Extract}_{\mathbb{P}}$ vanishes only at O , since 5 is a non-square in $\mathbb{F}_{p_{\text{Pallas}}}$; and $\text{Extract}_{\mathbb{P}}(P) = \text{Extract}_{\mathbb{P}}(P')$ if and only if $P' = \pm P$.*

The partial inverse $\text{abst}_{\mathbb{P}}$ rejects $\tilde{x} \geq p_{\text{Pallas}}$ and non-squares $x^3 + 5$: the star encoding is canonical.

$$P^* \in \{0, 1\}^{256}; \text{Extract}_{\mathbb{P}}: \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}) \rightarrow \mathbb{F}_{p_{\text{Pallas}}}.$$

Generators from a hash

The point $\text{GroupHash}(D, M) \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$: hash_to_field by expand_message_xmd over BLAKE2b-512 gives $u_0, u_1 \in \mathbb{F}_{p_{\text{Pallas}}}$; simplified SWU maps each to a curve 3-isogenous to Pallas; add; apply the isogeny.

$$\text{DST} := D \parallel \text{-pallas_XMD:BLAKE2b_SSWU_R0_}$$

(D, M)	generator	use
$(z.\text{cash}:\text{Orchard}, G)$	$G^{\text{SpendAuth}}$	spend authorisation
$(z.\text{cash}:\text{Orchard}, K)$	K^{Orchard}	nullifiers
$(z.\text{cash}:\text{Orchard-cv}, v), (\dots, r)$	$V^{\text{Orchard}}, R^{\text{Orchard}}$	value commitments
$(z.\text{cash}:\text{Orchard-gd}, d)$	g_d	diversified bases
$(z.\text{cash}:\text{SinsemillaS}, l2\text{LEOSP}_{32}(j))$	$S(j)$	Sinsemilla table

Assumption (random oracle). *Distinct pairs (D, M) receive independent uniform points.*

Primitives

Expansion and reduction

$$\text{PRFexpand}_k(t) := \text{BLAKE2b-512}(\text{Zcash_ExpandSeed}, k \parallel t) \in \{0, 1\}^{512}$$

$$\text{ToBase}(x) := \text{LEOS2IP}_{512}(x) \bmod p_{\text{Pallas}} \in \mathbb{F}_{p_{\text{Pallas}}},$$

$$\text{ToScalar}(x) := \text{LEOS2IP}_{512}(x) \bmod p_{\text{Vesta}} \in \mathbb{F}_{p_{\text{Vesta}}}$$

lead byte	key	derives, in order
06, 07, 08	sk	ask, nk, r_{ivk}
82	$\text{LE}_{256}(r_{\text{ivk}})$	dk $\parallel$ ovk
04, 09, 0B	rseed	esk, ψ , rcm

A 512-bit value reduced modulo a 255-bit prime is within statistical distance $m/2^{512} < 2^{-257}$ of uniform.

Assumption. The map $t \mapsto \text{PRFexpand}_k(t)$ is a PRF for uniform k ; distinct lead bytes give independent outputs.

The Sinsemilla hash

Chunk width $k = 10$; table

$S(j) := \text{GroupHash}(\text{z.cash:SinsemillaS}, \text{l2LEOSP}_{32}(j)), 0 \leq j < 2^{10}$; domain base

$Q(D) := \text{GroupHash}(\text{z.cash:SinsemillaQ}, D)$.

$$\text{Acc}_0 := Q(D), \quad \text{Acc}_i := (\text{Acc}_{i-1} \dot{+} S(m_i)) \dot{+} \text{Acc}_{i-1} \quad (1 \leq i \leq n)$$

$$\text{SinsemillaHashToPoint}(D, M) := \text{Acc}_n, \quad \text{SinsemillaHash}(D, M) := \text{Extract}_{\mathbb{P}}^{\perp}(\text{Acc}_n)$$

- The piece m_i is the i -th 10-bit piece of M ; at most $c = 253$ pieces, since $2^{253} \leq (p_{\text{Vesta}} - 1)/2$.
- Incomplete addition $\dot{+}$ returns $\perp$ when an operand is O or the x -coordinates agree.
- In a circuit each step is one table lookup and two additions.

Sinsemilla commitments and the three instances

$$\hat{M} := \text{SinsemillaHashToPoint}(D \parallel -M, M), \quad H_D := \text{GroupHash}(D \parallel -r, \varepsilon)$$

$$\text{SinsemillaCommit}_r(D, M) := \hat{M} + [r] H_D$$

$$\text{ShortCommit}_r(D, M) := \text{Extract}_{\mathbb{P}}^{\perp}(\text{SinsemillaCommit}_r(D, M))$$

$r \in \mathbb{F}_{p_{\text{Vesta}}}$; the commitment is $\perp$ when $\hat{M} = \perp$.

instance	form	message bits	chunks
NoteCommit	SinsemillaCommit	1086	109
Commit ^{ivk}	ShortCommit	510	51
MerkleCRH	SinsemillaHash	520	52

What Sinsemilla guarantees

Proposition (collision resistance). *Under the random-oracle model of GroupHash and discrete logarithms on Pallas, no efficient algorithm finds $M \neq M'$ of the domain's one fixed length l with $\text{SinsemillaHashToPoint}(D', M) = \pm \text{SinsemillaHashToPoint}(D', M') \neq \perp$, nor such a message on which $\text{SinsemillaHashToPoint}(D', \cdot) = \perp$. Each such output yields a non-trivial relation $\sum_i [a_i]P_i = O$ among $Q(D'), S(0), \dots, S(1023)$.*

Lemma (hiding and binding). (i) *If $H_D \neq O$ and r is uniform on $\mathbb{F}_{p_{\text{Vesta}}}$, then $\text{SinsemillaCommit}_r(D, M)$ is uniform on $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$ for every M with $\hat{M} \neq \perp$.* (ii) *Two openings, with $M \neq M'$ of one length l and commitments not $\perp$, of one commitment give a discrete-logarithm relation.*

Proof. For (i), the point $[r]H_D$ is uniform on a group of prime order. For (ii), the difference $\hat{M} - \hat{M}' = [r' - r]H_D$ expands into the generators. Fixed lengths matter: zero padding makes M and $M \parallel 0$ collide.

Poseidon and the nullifier PRF

Poseidon over $\mathbb{F}_{p_{\text{Pallas}}}$: width 3, rate 2, capacity 1, S-box $x \mapsto x^5$, 8 full and 56 partial rounds; $f: \mathbb{F}_{p_{\text{Pallas}}}^3 \rightarrow \mathbb{F}_{p_{\text{Pallas}}}^3$ the permutation.

$$\text{PoseidonHash}(x, y) := \pi_1(f(x, y, 2^{65}))$$

$$\text{PRF}_{\text{nk}}^{\text{nfOrchard}}(\rho) := \text{PoseidonHash}(\text{nk}, \rho)$$

$\text{nk}, \rho \in \mathbb{F}_{p_{\text{Pallas}}}$; the capacity element 2^{65} encodes the input length 2.

Assumption. For uniform $\text{nk}, \rho \mapsto \text{PoseidonHash}(\text{nk}, \rho)$ is a PRF $\mathbb{F}_{p_{\text{Pallas}}} \rightarrow \mathbb{F}_{p_{\text{Pallas}}}$.

Its status is cryptanalytic, with no reduction to a standard problem; only nullifier unlinkability uses it.

RedPallas

Base $B \neq O$; hash $H^\otimes(z) := \text{ToScalar}(\text{BLAKE2b-512}(\text{Zcash_RedPallasH}, z)); \underline{P} :=$
the 32 bytes of P^* .

keys $x \in \mathbb{F}_{p_{\text{Vesta}}}, \quad X := [x] B$

sign $M \quad T \leftarrow \{0, 1\}^{640}, \quad r := H^\otimes(T \parallel \underline{X} \parallel M), \quad R := [r] B$
 $c := H^\otimes(\underline{R} \parallel \underline{X} \parallel M), \quad S := r + cx, \quad \sigma := \underline{R} \parallel \text{I2LEOSP}_{256}(S)$

verify $R \neq \perp, \quad S < p_{\text{Vesta}}, \quad [S] B = R + [c] X$

instance	base B	key re-randomisation
spend authorisation	$G^{\text{SpendAuth}}$	yes: $(x + \alpha, X + [\alpha] B)$
binding signature	R^{Orchard}	no

Keys and addresses

The spend-side secrets

$$\text{ask} := \text{ToScalar}(\text{PRFexpand}_{\text{sk}}([06])) \in \mathbb{F}_{p_{\text{Vesta}}}$$

$$\text{nk} := \text{ToBase}(\text{PRFexpand}_{\text{sk}}([07])) \in \mathbb{F}_{p_{\text{Pallas}}}$$

$$\text{r}_{\text{ivk}} := \text{ToScalar}(\text{PRFexpand}_{\text{sk}}([08])) \in \mathbb{F}_{p_{\text{Vesta}}}$$

$$\text{ak}^{\mathbb{P}} := [\text{ask}] G^{\text{SpendAuth}} \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}), \quad \text{ak} := \text{Extract}_{\mathbb{P}}(\text{ak}^{\mathbb{P}}) \in \mathbb{F}_{p_{\text{Pallas}}}$$

Sign normalisation: if $[\text{ask}] G^{\text{SpendAuth}}$ has odd y , replace ask by $-\text{ask}$. The point $\text{ak}^{\mathbb{P}}$ is recovered from ak by the even- y lift.

$\text{sk} \in \{0, 1\}^{256}$ uniform; key generation redraws sk if $\text{ask} = 0$, $\text{ivk} \in \{0, \perp\}$ or the ZIP 32 internal key's $\text{ivk} \in \{0, \perp\}$.

Viewing keys

$$\text{fvk} := (\text{ak}, \text{nk}, r_{\text{ivk}})$$

$$\text{ivk} := \text{Commit}_{r_{\text{ivk}}}^{\text{ivk}}(\text{ak}, \text{nk}) = \text{Extract}_{\mathbb{P}}(\hat{M} + [r_{\text{ivk}}]H_D),$$

$$\hat{M} := \text{SinsemillaHashToPoint}(D \parallel -M, \text{LE}_{255}(\text{ak}) \parallel \text{LE}_{255}(\text{nk})),$$

$$\text{dk} \parallel \text{ovk} := \text{PRFexpand}_{\text{LE}_{256}(r_{\text{ivk}})}([82] \parallel \text{I2LEOSP}_{256}(\text{ak}) \parallel \text{I2LEOSP}_{256}(\text{nk}))$$

$D := \text{z.cash:Orchard-CommitIvk}; \text{ivk} \in \{1, \dots, p_{\text{Pallas}} - 1\} \subset \mathbb{F}_{p_{\text{Vesta}}}$ for an accepted key;
 dk, ovk 32 bytes each.

Lemma. For r_{ivk} uniform and independent of (ak, nk) with $\hat{M} \neq \perp$, ivk is 0 with probability $1/p_{\text{Vesta}}$ and otherwise uniform on the $(p_{\text{Vesta}} - 1)/2$ x-coordinates of non-identity points.

Proposition (binding). No efficient algorithm finds $(\text{ak}, \text{nk}, r_{\text{ivk}}) \neq (\text{ak}', \text{nk}', r_{\text{ivk}}')$ with equal $\text{Commit}^{\text{ivk}} \neq \perp$.

Diversified addresses

For an index $j \in \{0, \dots, 2^{88} - 1\}$:

$$d := \text{FF1-AES256}_{\text{dk}}(\text{LE}_{88}(j)) \in \{0, 1\}^{88}$$

$$g_d := \text{DiversifyHash}(d) := T \text{ if } T \neq O, \text{ else } \text{GroupHash}(z.\text{cash}:\text{Orchard-gd}, \varepsilon),$$

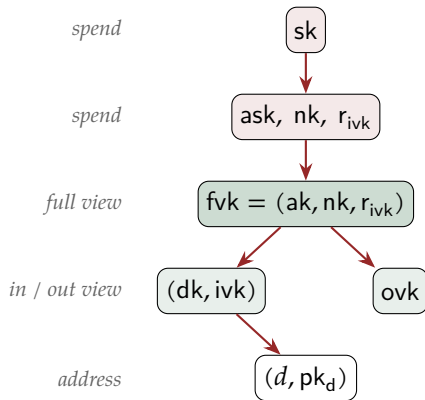
where $T := \text{GroupHash}(z.\text{cash}:\text{Orchard-gd}, d)$

$$\text{pk}_d := [\text{ivk}] g_d$$

The address is (d, pk_d) , with $\text{pk}_d \neq O$. One key has 2^{88} addresses, all with transmission keys under the same ivk .

Proposition (unlinkability). *For keys with $\text{use_qsk} = \text{false}$, given ak , nk and ovk of two keys, and addresses of both, no efficient adversary tells which key a further address at a fresh index belongs to. Assumptions: FF1 a PRP, DDH on Pallas, joint hiding of the viewing keys, PRF expansion.*

The key ladder



ask, nk, r_{ivk} from $\text{PRFexpand}_{sk}([06]), ([07]), ([08])$

$ak^{\mathbb{P}} := [ask] G^{\text{SpendAuth}}, \quad ak := \text{Extract}_{\mathbb{P}}(ak^{\mathbb{P}})$

$ivk := \text{Commit}_{r_{ivk}}^{ivk}(ak, nk)$

$dk \parallel ovk := \text{PRFexpand}_{r_{ivk}}([82] \parallel ak \parallel nk)$

$d := \text{FF1}_{dk}(\text{LE}_{88}(j)), \quad g_d := \text{DiversifyHash}(d)$

$pk_d := [ivk] g_d$

Every arrow is one-way: each lower key is computed from the one above, and not conversely.

What each key can do

Proposition (capability separation). *For an honestly generated key with $\text{use_qsk} = \text{false}$, no efficient algorithm, given only*

- (a) the full viewing key $(\text{ak}, \text{nk}, \text{r}_{\text{ivk}})$, outputs ask;
- (b) the incoming viewing key (dk, ivk) , outputs nk or ask;
- (c) ovk, outputs ivk;
- (d) an address (d, pk_d) , outputs ivk.

key	computes	cannot
sk, ask	signatures: spend	—
fvk	nullifiers: detect spends	sign
(dk, ivk)	decrypt received notes, derive addresses	detect spends
ovk	recover sent notes	derive ivk

Notes and the tree

The note

$$n = ((d, \text{pk}_d), v, \rho, \psi, \text{rcm})$$

field	type	source
(d, pk_d)	$\{0, 1\}^{88} \times (\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}) \setminus \{O\})$	the recipient's address
v	$\{0, \dots, 2^{64} - 1\}$ zatoshi	the amount; 1 ZEC = 10^8
ρ	$\mathbb{F}_{p_{\text{Pallas}}}$	the nullifier consumed in the same Action
ψ	$\mathbb{F}_{p_{\text{Pallas}}}$	derived from the note seed
rcm	$\mathbb{F}_{p_{\text{Vesta}}}$	derived from the note seed

No note appears on the chain in the clear.

The note commitment

$$M(n) := g_d^* \parallel pk_d^* \parallel LE_{64}(v) \parallel LE_{255}(\rho) \parallel LE_{255}(\psi) \quad (1086 \text{ bits})$$

$$cm := \text{NoteCommit}_{rcm}(M(n)) = \hat{M} + [rcm]H_D, \quad cmx := \text{Extract}_{\mathbb{P}}^{\perp}(cm)$$

$\hat{M}$ under `z.cash:Orchard-NoteCommit-M`;

$H_D := \text{GroupHash}(\text{z.cash:Orchard-NoteCommit-r}, \varepsilon)$.

Proposition. (a) *Perfect hiding:* for rcm uniform, cm is uniform on $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$ for every note content with $\hat{M} \neq \perp$. (b) *Binding:* no efficient algorithm outputs notes $n \neq n'$ with $cmx(n) = cmx(n') \neq \perp$.

Proof. Part (a) is the Lemma on `SinsemillaCommit`. For (b), distinct d with equal g_d is a `GroupHash` relation; otherwise equal extracted values mean $cm' = \pm cm$, and either sign gives a relation among the `Sinsemilla` generators and H_D .

The note seed

The sender draws $rseed \leftarrow \{0, 1\}^{256}$; with $\underline{\rho} := I2LEOSP_{256}(\rho)$ and lead byte 0x03:

$$esk := \text{ToScalar}(\text{PRFexpand}_{rseed}([04] \parallel \underline{\rho}))$$

$$\psi := \text{ToBase}(\text{PRFexpand}_{rseed}([09] \parallel \underline{\rho}))$$

$$rcm := \text{ToScalar}(\text{PRFexpand}_{rseed}(\text{pre_rcm}))$$

$$\text{pre_rcm} := [0B] \parallel g_d^* \parallel pk_d^* \parallel I2LEOSP_{64}(v) \parallel \underline{\rho} \parallel I2LEOSP_{256}(\psi) \quad (137 \text{ bytes})$$

Redraw $rseed$ if $esk = 0$ or $cmx = \perp$. One seed fixes the note's randomness and the encryption key; a recipient can re-derive all three.

The note commitment tree

Depth 32; layer 32 holds the leaves, layer 0 the root.

$$M_i^{32} := \begin{cases} \text{cmx appended at position } i & i < n \\ \text{Uncommitted} = 2 & n \leq i < 2^{32} \end{cases}$$

$$M_i^h := \text{MerkleCRH}(h, M_{2i}^{h+1}, M_{2i+1}^{h+1}) \quad (0 \leq h < 32)$$

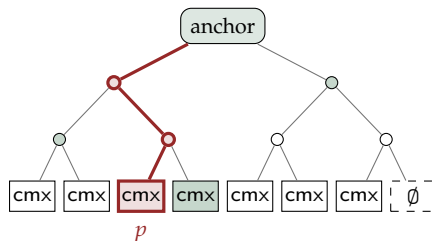
$$\text{MerkleCRH}(l, x, y) := \text{SinsemillaHash}(D, \text{LE}_{10}(31 - l) \parallel \text{LE}_{255}(x) \parallel \text{LE}_{255}(y))$$

$D := \text{z.cash:Orchard-MerkleCRH}$; an output $\perp$ is replaced by 0; messages of 520 bits.

The *anchor* of a block is the root M_0^0 of its final treestate.

Lemma. No point P has $\text{Extract}_{\mathbb{P}}(P) = 2$, since $2^3 + 5 = 13$ is a non-square in $\mathbb{F}_{p_{\text{Pallas}}}$: an unused leaf is the commitment of no note.

Authentication paths



Write $\text{pos} = \sum_{t < 32} b_t 2^t$, siblings $s_0, \dots, s_{31}$, and $h_t := \text{MerkleCRH}(31 - t, \cdot, \cdot)$:

$$a_0 := v, \quad a_{t+1} := \begin{cases} h_t(a_t, s_t) & b_t = 0 \\ h_t(s_t, a_t) & b_t = 1 \end{cases}$$

Valid path to rt : $a_{32} = rt$.

Every sibling s_t is a function of the leaves published through the anchoring block, so the holder computes the path from public data and never sends pos.

Membership soundness

Proposition. *Let rt be an anchor whose tree holds n leaves $L_0, \dots, L_{n-1}$. From a valid path to rt with $v \neq L_{\text{pos}}$ one computes either two distinct 520-bit messages with a common 10-bit prefix and equal SinsemillaHash, or a message on which SinsemillaHash returns $\perp$. Hence no efficient adversary, even one that chooses the leaves, produces such a path; and if $v = \text{Extract}_{\mathbb{P}}(P)$ for a point P , then $\text{pos} < n$ and v is the leaf appended at pos .*

Proof. walk the two folds from the leaf upwards; the first level at which they agree after differing yields the collision, and the level tag $\text{LE}_{10}(t)$ of step t makes the two messages share a prefix.

Nullifiers

The nullifier

$$\begin{aligned}\text{DeriveNullifier}_{\text{nk}}(\rho, \psi, \text{cm}) &:= \text{Extract}_{\mathbb{P}}([s] K^{\text{Orchard}} + \text{cm}), \\ s &:= (\text{PRF}_{\text{nk}}^{\text{nfOrchard}}(\rho) + \psi) \bmod p_{\text{Pallas}}\end{aligned}$$

$\text{nk}, \rho, \psi \in \mathbb{F}_{p_{\text{Pallas}}}; \text{cm} \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}); K^{\text{Orchard}} := \text{GroupHash}(\text{z.cash:Orchard}, K).$

- Deterministic: one note under one key has one nullifier.
- Keyed: without nk the multiplier s is pseudorandom.
- Bound to the commitment: cm enters additively.

Chaining. In every Action, $\rho^{\text{new}} := \text{nf}^{\text{old}}$. Accepted Actions of one pool publish distinct nullifiers (the nullifier-set rule), so within a pool no two created notes share ρ .

Uniqueness and unlinkability

Proposition (uniqueness). *No efficient adversary, holding every key, outputs notes with commitments $cm \neq cm'$, not $\perp$, and keys nk, nk' with equal nullifiers.*

Proof. equal x -coordinates give $[s] K^{\text{Orchard}} + cm = \pm([s'] K^{\text{Orchard}} + cm')$; expanding the commitments gives a relation among K^{Orchard} , H_D and the Sinsemilla generators.

Proposition (unlinkability). *For a key with $\text{use_qsk} = \text{false}$, given (dk, ivk) and $ak^{\mathbb{P}}$ but not nk , an adversary that submits q notes with distinct ρ_i cannot tell nf_i from $\text{Extract}_{\mathbb{P}}([u_i] K^{\text{Orchard}} + cm_i)$ with u_i uniform, up to $\epsilon_{\text{Pos}} + q \delta$ plus key-generation terms, where $\delta = (p_{\text{Vesta}} - p_{\text{Pallas}})/p_{\text{Vesta}} < 2^{-167}$.*

Value

Value commitments

$$V^{\text{Orchard}} := \text{GroupHash}(z.\text{cash}:\text{Orchard-cv}, v),$$

$$R^{\text{Orchard}} := \text{GroupHash}(z.\text{cash}:\text{Orchard-cv}, r)$$

$$v^{\text{net}} := v^{\text{old}} - v^{\text{new}} \in \{-2^{64} + 1, \dots, 2^{64} - 1\}$$

$$\text{cv}^{\text{net}} := \text{ValueCommit}_{\text{rcv}}(v^{\text{net}}) := [v^{\text{net}} \bmod p_{\text{Vesta}}] V^{\text{Orchard}} + [\text{rcv}] R^{\text{Orchard}}$$

$\text{rcv} \in \mathbb{F}_{p_{\text{Vesta}}}$ drawn uniformly per Action, from no note; $\text{cv}^{\text{net}} \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$.

Lemma. (i) *Perfect hiding:* for uniform rcv , cv^{net} is uniform on $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$. (ii) *Binding:* two openings with $v \not\equiv v' \pmod{p_{\text{Vesta}}}$ give $\log_{R^{\text{Orchard}}} V^{\text{Orchard}}$.

The binding signature

Balancing value $v^{\text{balance}} \in \{-2^{63}, \dots, 2^{63} - 1\}$: net value leaving the pool, public.

$$\text{bsk} := \sum_{i=1}^n \text{rcv}_i, \quad \text{bvk} := \sum_{i=1}^n \text{cv}_i^{\text{net}} - [v^{\text{balance}}] V^{\text{Orchard}}$$

$$\text{bvk} - [\text{bsk}] R^{\text{Orchard}} = \left[\sum_i v_i^{\text{net}} - v^{\text{balance}} \right] V^{\text{Orchard}}$$

Proposition. (i) The key satisfies $\text{bvk} = [\text{bsk}] R^{\text{Orchard}}$ if and only if $\sum_i v_i^{\text{net}} \equiv v^{\text{balance}} \pmod{p_{\text{Vesta}}}$. (ii) A party that opens every cv_i^{net} and knows b with $\text{bvk} = [b] R^{\text{Orchard}}$, while $D := \sum_i v'_i - v^{\text{balance}} \not\equiv 0$, knows $\log_{R^{\text{Orchard}}} V^{\text{Orchard}} = (b - \sum_i r'_i) D^{-1}$.

A RedPallas signature under bvk on base R^{Orchard} proves knowledge of bsk .

Modular balance is integer balance

Lemma. *Let $n < 2^{16}$, $v_i^{\text{old}}, v_i^{\text{new}} \in \{0, \dots, 2^{64} - 1\}$ and $|v^{\text{balance}}| \leq \text{MAX_MONEY}$. If $\sum_i (v_i^{\text{old}} - v_i^{\text{new}}) \equiv v^{\text{balance}} \pmod{p_{\text{Vesta}}}$, the two sides are equal as integers.*

Proof. their difference has absolute value at most

$$(2^{16} - 1)(2^{64} - 1) + \text{MAX_MONEY} = 1\,208\,907\,374\,970\,555\,465\,089\,025 < 2^{80} < p_{\text{Vesta}}.$$

The action count and the range of v^{balance} are consensus rules; the 64-bit ranges of $v^{\text{old}}, v^{\text{new}}$ are enforced inside the proof.

Spend authorisation

Randomised validating keys

Per Action the spender draws $\alpha := H^{\otimes}(T)$, $T \leftarrow \{0, 1\}^{640}$:

$$\text{rsk} := \text{ask} + \alpha, \quad \text{rk} := \text{ak}^{\mathbb{P}} + [\alpha] G^{\text{SpendAuth}} = [\text{rsk}] G^{\text{SpendAuth}}$$

The Action publishes rk and a signature under rsk on SigHash; never ask, α , $\text{ak}^{\mathbb{P}}$ or ak.

Proposition (unlinkability). *For α_i uniform and independent, $(\text{rk}_1, \dots, \text{rk}_n)$ is uniform on $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})^n$ whatever the keys $\text{ak}_i^{\mathbb{P}}$: two Actions of one key and of two keys are identically distributed. With α_i from $H^{\otimes}$ the distance is at most $2n(\delta_{512} + (q_h + n) 2^{-640})$, $\delta_{512} < 2^{-257}$.*

Unforgeability under re-randomisation

Proposition. Let $\text{ak}^{\mathbb{P}} := [x] G^{\text{SpendAuth}}$ for uniform x . An adversary with q_h hash queries and q_s signing queries (α, M) , answered under $x + \alpha$, outputs $(\text{rk}^*, M^*, \sigma^*)$ and α^* with $\text{rk}^* = \text{ak}^{\mathbb{P}} + [\alpha^*] G^{\text{SpendAuth}}$. If it forges with probability ϵ , an algorithm without x outputs x with probability at least

$$\frac{(\epsilon - \eta)^2}{Q} - \frac{Q}{p_{\text{Vesta}}} - \frac{q_s(q_h + q_s)}{p_{\text{Vesta}}}, \quad Q := q_h + q_s + 1,$$
$$\eta := Q \delta_{512} + q_s(q_h + 2q_s) 2^{-640}.$$

Proof. simulate signatures by programming H ; fork the adversary at the challenge query; two forgeries with distinct challenges give rsk^* , and $x = \text{rsk}^* - \alpha^*$.

With every $\alpha = 0$ this is strong unforgeability of RedPallas.

The Action statement

Bundle flags

One flags byte $f = \sum_{i=0}^7 f_i 2^i$ per bundle:

$$\text{enableSpend} := f_0, \quad \text{enableOutput} := f_1$$

$$\text{enableCrossAddress} := \begin{cases} f_2 & \text{in the Ironwood pool} \\ 0 & \text{in the Orchard pool} \end{cases}$$

$$\delta := \text{disableCrossAddress} := 1 - \text{enableCrossAddress}$$

- In `flagsIronwood` bits 3–7 are reserved and 0; in `flagsOrchard` bits 2–7 are.
- The *expanded receiver* of an address (d, pk_d) is (g_d, pk_d) .
- With $\delta = 1$ an Action may only create its note at the expanded receiver of the note it consumes. Every Orchard-pool Action has $\delta = 1$ (ZIP 258), which disables cross-address transfers; no value enters that pool, by the separate rule $\text{valueBalanceOrchard} \geq 0$.

The relation: inputs and types

The relation $\mathcal{R}_{\text{Action}}$ is the set of pairs (pub, aux) with

$$\text{pub} = (rt, cv^{\text{net}}, nf^{\text{old}}, rk, cmx, \text{enableSpend}, \text{enableOutputs}, \delta)$$

$$\text{aux} = (\text{path}, \text{pos}, g_d^{\text{old}}, pk_d^{\text{old}}, v^{\text{old}}, \rho^{\text{old}}, \psi^{\text{old}}, rcm^{\text{old}}, cm^{\text{old}}, \alpha, ak^{\mathbb{P}}, nk, r_{ivk}, \\ g_d^{\text{new}}, pk_d^{\text{new}}, v^{\text{new}}, \psi^{\text{new}}, rcm^{\text{new}}, rcv, c)$$

$$rt, nf^{\text{old}}, cmx \in \{0, \dots, p_{\text{Pallas}} - 1\}$$

$$g_d^{\text{old}}, pk_d^{\text{old}}, g_d^{\text{new}}, pk_d^{\text{new}}, ak^{\mathbb{P}} \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}}) \setminus \{O\}$$

$$\rho^{\text{old}}, \psi^{\text{old}}, \psi^{\text{new}}, nk \in \mathbb{F}_{p_{\text{Pallas}}}$$

$$\text{path} \in \{0, \dots, p_{\text{Pallas}} - 1\}^{32}, \text{pos} < 2^{32}$$

$$cv^{\text{net}}, rk, cm^{\text{old}} \in \mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$$

$$v^{\text{old}}, v^{\text{new}} \in \{0, \dots, 2^{64} - 1\}$$

$$rcm^{\text{old}}, rcm^{\text{new}}, r_{ivk}, rcv, \alpha < 2^{255}$$

$$c \in \{0, 1\}^{64}$$

Set $\rho^{\text{new}} := nf^{\text{old}}$; $M^{\text{old}}, M^{\text{new}}$ are the 1086-bit note messages.

Conditions A1 to A5

- A1. $\text{NoteCommit}_{\text{rcm}^{\text{old}}}(M^{\text{old}}) \in \{\text{cm}^{\text{old}}, \perp\}$ old note commitment integrity
- A2. $\text{Extract}_{\mathbb{P}}^{\perp}(\text{NoteCommit}_{\text{rcm}^{\text{new}}}(M^{\text{new}})) \in \{\text{cmx}, \perp\}$ new note commitment integrity
- A3. $v^{\text{old}} \cdot (\text{root} - \text{rt}) = 0$, or a hash in the fold returns $\perp$ Merkle path validity
 $\text{root} := a_{32}$, the fold from $a_0 := \text{Extract}_{\mathbb{P}}(\text{cm}^{\text{old}})$ along $(\text{pos}, \text{path})$
- A4. $\text{cv}^{\text{net}} = [v^{\text{old}} - v^{\text{new}}] V^{\text{Orchard}} + [\text{rcv}] R^{\text{Orchard}}$ value commitment integrity
- A5. $\text{nf}^{\text{old}} = \text{Extract}_{\mathbb{P}}([(\text{PRF}_{\text{nk}}^{\text{nfOrchard}}(\rho^{\text{old}}) + \psi^{\text{old}}) \bmod p_{\text{Pallas}}] K^{\text{Orchard}} + \text{cm}^{\text{old}})$ nullifier integrity

The bits c admit the non-canonical encodings $x + p_{\text{Pallas}} < 2^{255}$ of the fold's arguments; the values $\perp$ cannot be produced efficiently.

Conditions A6 to A10

A6. $rk = ak^{\mathbb{P}} + [\alpha] G^{\text{SpendAuth}}$ spend authority

A7. $ivk := \text{Commit}_{r_{ivk}}^{ivk}(\text{Extract}_{\mathbb{P}}(ak^{\mathbb{P}}), nk); ivk = \perp \text{ or } pk_d^{\text{old}} = [ivk] g_d^{\text{old}}$
diversified address integrity

A8. $v^{\text{old}} \cdot (1 - \text{enableSpend}) = 0$ enable spends

A9. $v^{\text{new}} \cdot (1 - \text{enableOutput}) = 0$ enable outputs

A10. $\delta(x(g_d^{\text{old}}) - x(g_d^{\text{new}})) = \delta(y(g_d^{\text{old}}) - y(g_d^{\text{new}})) = 0,$
 $\delta(x(pk_d^{\text{old}}) - x(pk_d^{\text{new}})) = \delta(y(pk_d^{\text{old}}) - y(pk_d^{\text{new}})) = 0$ cross-address restriction

Every condition is an equation over $\mathbb{F}_{p_{\text{Pallas}}}$ or $\mathcal{E}(\mathbb{F}_{p_{\text{Pallas}}})$: exactly the shape a circuit over $\mathbb{F}_{p_{\text{Pallas}}}$ checks.

What the statement enforces, and what is checked outside

requirement	inside the proof	outside the proof
R1 hidden value	A1, A2	—
R2 membership	A1, A3	the anchor is a past root
R3 no double spend	A5, chaining $\rho^{\text{new}} = \text{nf}^{\text{old}}$	nf not in the set
R4 conservation	A4, the 64-bit ranges	binding signature under bvk
R5 authority	A6, A7	signature under rk on SigHash
flags, seal	A8, A9, A10	flags read from the bundle

Not checked by the statement: the derivation of rcm and ψ from rseed, and the note ciphertexts. The recipient checks those.

The proof system, as assumptions

Assumption (knowledge soundness). *With the Fiat–Shamir hash a random oracle, for every efficient prover that outputs primary inputs $\text{pub}_1, \dots, \text{pub}_n$ and an accepted proof there is an efficient extractor outputting $\text{aux}_1, \dots, \text{aux}_n$ with $(\text{pub}_i, \text{aux}_i) \in \mathcal{R}_{\text{Action}}$ for all i , except with negligible probability.*

Assumption (zero knowledge). *With a programmable random oracle, a simulator given only $\text{pub}_1, \dots, \text{pub}_n$ outputs proofs statistically indistinguishable from honest ones.*

One proof per bundle, over n instances of one circuit; the last part of this deck shows how Halo 2 supports both.

Note encryption

The note plaintext

$\text{np} := \text{leadByte} \parallel d \parallel \text{I2LEOSP}_{64}(v) \parallel \text{rseed} \parallel \text{memo}$

field	bytes	
leadByte	1	0x03 in the Ironwood pool
d	11	the diversifier
v	8	the value
rseed	32	the note seed
memo	512	chosen by the sender
total	564	fixed: the length reveals nothing

From (d, v, rseed) and the Action's nf^{old} the recipient recomputes g_d , ψ , rcm and esk .

Encryption to the recipient

$$\text{epk} := [\text{esk}] g_d, \quad s := [\text{esk}] \text{pk}_d$$

$$K_{\text{enc}} := \text{BLAKE2b-256}(\text{Zcash_OrchardKDF}, s^* \parallel \text{epk}^*)$$

$$C_{\text{enc}} := \text{ChaCha20-Poly1305}_{K_{\text{enc}}}(\text{np}) \quad (\text{zero nonce, } 564 + 16 = 580 \text{ bytes})$$

Proposition (correctness). *If $\text{pk}_d = [\text{ivk}] g_d$ then $[\text{ivk}] \text{epk} = [\text{esk}] \text{pk}_d = s$: the holder of ivk derives the same key.*

Proposition (confidentiality and key privacy). *For an honestly generated recipient with $\text{use_qsk} = \text{false}$, $(\text{epk}^*, C_{\text{enc}}, \text{cmx})$ is computationally indistinguishable from a uniform non-identity point, an encryption of a fixed string under an independent key, and cmx. DDH on Pallas, KDF a random oracle, the AEAD, joint hiding of the viewing keys.*

The outgoing ciphertext

$$\text{ock} := \text{BLAKE2b-256}(\text{Zcash_Orchardock}, \text{ovk} \parallel (\text{cv}^{\text{net}})^* \parallel \text{I2LEOSP}_{256}(\text{cmx}) \parallel \text{epk}^*)$$

$$\text{op} := \text{pk}_d^* \parallel \text{I2LEOSP}_{256}(\text{esk}) \quad (64 \text{ bytes})$$

$$C_{\text{out}} := \text{ChaCha20-Poly1305}_{\text{ock}}(\text{op}) \quad (80 \text{ bytes})$$

The sender's ovk recovers $(\text{pk}_d, \text{esk})$, hence $s = [\text{esk}] \text{pk}_d$ and the note: a wallet reconstructs what it sent. With $\text{ovk} = \perp$ both ock and op are random and nothing is recoverable.

Trial decryption and acceptance

For every Action: $s := [\text{ivk}] \text{epk}$, $K_{\text{enc}} := \text{KDF}(s, \text{epk})$, $\text{np} := \text{Dec}_{K_{\text{enc}}}(C_{\text{enc}})$; a result $\neq \perp$ is a candidate. For an Ironwood-pool output, accept only if:

- (i) $\text{leadByte} = 0\text{x}03$;
- (ii) with $\rho := \text{nf}^{\text{old}}$: $\text{g}_d := \text{DiversifyHash}(d)$, $\text{pk}_d := [\text{ivk}] \text{g}_d$, then ψ , then rcm from rseed ;
- (iii) $([\text{esk}] \text{g}_d)^* = \text{epk}^*$ for esk from rseed ;
- (iv) $\text{Extract}_{\mathbb{P}}(\text{NoteCommit}_{\text{rcm}}(M(n))) = \text{cmx}$.

Proposition. *An accepted note is addressed to the recipient's key, carries $\rho = \text{nf}^{\text{old}}$, and is the unique opening of cmx : no sender can make the recipient accept n while holding $n' \neq n$ with $\text{cmx}(n') = \text{cmx}$.*

Transactions and consensus

The Action description

field	bytes	
$(cv^{\text{net}})^*$	32	net value commitment
$I2LEOSP_{256}(nf)$	32	nullifier of the consumed note
rk^*	32	randomised validating key
$I2LEOSP_{256}(cmx)$	32	extracted commitment of the created note
epk^*	32	ephemeral public key
C_{enc}	580	note ciphertext
C_{out}	80	outgoing ciphertext
	820	

Per bundle: flags (1), v^{balance} (8), anchor (32), proof ($2720 + 2272n$), n spend signatures and one binding signature (64 each).

Digests: what the signatures sign

$$\text{txid} := h(\text{ZcashTxHash_} \parallel \beta, d_{\text{hdr}} \parallel d_{\text{tr}} \parallel d_{\text{Sap}} \parallel d_{\text{Orc}} \parallel d_{\text{lrw}})$$

$$d_{\text{lrw}} := h(\text{ZTxIdIronwd_H_v6}, d_{\text{C}} \parallel d_{\text{M}} \parallel d_{\text{N}} \parallel \text{flags} \parallel v^{\text{balance}})$$

The hash h is BLAKE2b-256 under a 16-byte personalisation; β the consensus branch identifier; $d_{\text{C}}, d_{\text{M}}, d_{\text{N}}$ hash the Actions' compact, memo and remaining fields.

For a transaction without transparent inputs the signature digest SigHash equals txid; otherwise its transparent child also commits to the spent coins' values and scripts. It is the message of every spend-authorisation signature and of each binding signature.

Proposition (bundle binding). *Under collision resistance of BLAKE2b, discrete logarithms on Pallas and H a random oracle, no efficient adversary, against honestly generated keys whose holders sign only $\text{SigHash}(T)$ under each rk , produces an accepted transaction that reuses an Action's rk and changes the effecting data, except with negligible probability. The anchor and the proof are not signed: they are authorising data, and the proof binds the anchor.*

Verification

A version-5 or version-6 transaction is accepted, for its Orchard-protocol bundles, iff:

1. every field parses canonically; counts, reserved bits, flags, ranges;
2. the proof verifies on the n primary inputs $(rt, cv_i^{\text{net}}, nf_i, rk_i, cmx_i, f_0, f_1, \delta)$ R1–R5 via A1–A10
3. rt is the root of the pool's tree at an earlier block R2
4. no nullifier repeats, in the transaction or the pool's set R3
5. each spend signature verifies under rk_i on SigHash R5
6. the binding signature verifies under bvk ; the fee is ≥ 0 R4
7. pool rules: expiry, tree capacity, chain value pool balances.

On acceptance: nullifiers enter the pool's set, each cmx is appended to its tree, and the pool's balance decreases by v^{balance} .

As specified for NU7

- Block target spacing 25 s (ZIP 218); the default expiry delta is 120 blocks, 3000 s = 50 minutes; an anchor depth of 3 blocks is 75 s.
- Version-4 transactions are invalid (ZIP 2003); versions 5 and 6 remain, with no new format (ZIP 259).
- Per-block shielded limits (ZIP 218): `OrchardProtocolBlockActionLimit` = 330 per pool; Orchard-pool and Ironwood-pool Actions share one `GlobalShieldedBudget` = 330.
- The Orchard pool stays sealed, as from NU6.3 (ZIP 258): no Orchard-pool Actions in coinbase, $\delta = 1$ in every Orchard-pool Action, and `valueBalanceOrchard` ≥ 0 .

ZIP 259, which deploys NU7, has status Draft.

A worked payment

Alice holds a note of 7 ZEC in the Ironwood pool and pays Bob 5 ZEC.

Action	v^{old}	v^{new}	v^{net}
1: Alice's note $\rightarrow$ Bob	700 000 000	500 000 000	200 000 000
2: dummy $\rightarrow$ Alice's change	0	199 990 000	-199 990 000
$v^{\text{balance}} = \sum v^{\text{net}}$			10 000

- Fee $5000 \cdot \max(2, 2) = 10\,000$ zatoshi (ZIP 317), paid by $v^{\text{balance}} > 0$ into the transparent value pool.
- The binding key is $\text{bvk} = \sum c v_i^{\text{net}} - [10\,000] V^{\text{Orchard}} = [\text{rcv}_1 + \text{rcv}_2] R^{\text{Orchard}}$.
- The flag `enableCrossAddress` is 1: Action 1 pays a receiver other than its consumed note's.
- The Ironwood component: 9141 bytes, of which the proof 7264.

Security

Assumptions, listed once

assumption	used for
discrete logarithms on Pallas	every binding and uniqueness result
GroupHash a random oracle	independent generators
knowledge soundness of the Action proof	every soundness theorem
zero knowledge of the Action proof	privacy
PRFexpand a PRF	keys, note seeds
Poseidon a PRF	nullifier unlinkability
DDH on Pallas; FF1 a PRP	note encryption; address unlinkability
joint hiding of the viewing keys	capabilities, privacy
H, KDF, PRF^{ock} random oracles	signatures, encryption
ChaCha20-Poly1305 IND-CPA, INT-CTXT	note encryption
BLAKE2b collision resistance	bundle binding

Each theorem names the ones it uses; none is a theorem itself.

Balance

Theorem. *For each pool, every accepted bundle output by an efficient adversary satisfies $\sum_{i=1}^n (v_i^{\text{old}} - v_i^{\text{new}}) = v^{\text{balance}}$ as integers, the values read from the extracted witnesses, except with negligible probability.*

Proof. Extract the witnesses (knowledge soundness). By A4 each cv_i^{net} opens to $(v_i^{\text{old}} - v_i^{\text{new}}, \text{rcv}_i)$. The binding signature verifies under bvk , so a key b with $\text{bvk} = [b] R^{\text{Orchard}}$ is extracted by forking. If $\sum_i v_i^{\text{net}} \neq v^{\text{balance}}$, part (ii) of the binding proposition gives $\log_{R^{\text{Orchard}}} V^{\text{Orchard}}$, a discrete-logarithm relation between two GroupHash outputs. So the sum holds modulo p_{Vesta} , hence over the integers by the 2^{80} bound.

No double-spending

Lemma. *Two accepted Actions whose witnessed consumed notes have the same $\text{Extract}_{\mathbb{P}}(\text{cm}^{\text{old}})$ consume the same note under the same nk and publish the same nf^{old} .*

Proof. Condition A1 and binding of NoteCommit fix the note, hence $(g_d^{\text{old}}, pk_d^{\text{old}})$; by A7, $pk_d^{\text{old}} = [\text{ivk}] g_d^{\text{old}}$ fixes ivk , and binding of Commit^{ivk} fixes (ak, nk) ; A5 is then determined.

Theorem. *For each pool, no committed note of non-zero value is consumed by two accepted Actions of that pool, except with negligible probability.*

Proof. Membership (A3) makes each consumed note a leaf; the lemma gives one nullifier; the nullifier rule rejects the second.

Corollary (Faerie Gold). *Distinct accepted Actions of one pool create notes with distinct ρ ; distinct notes have distinct nullifiers.*

Spend authority

Experiment. *The adversary gets the full viewing key (ak, nk, r_{ivk}) , not ask, and signatures on queries (α, m) under $ask + \alpha$. It wins with an accepted Action that consumes a note of the key and carries a valid signature under its rk on some m with no query (α, m) satisfying $rk = ak^{\mathbb{P}} + [\alpha] G^{\text{SpendAuth}}$.*

Theorem. *No efficient adversary wins, except with negligible probability. Keys with $use_qsk = \text{false}$.*

Proof. Extract the witness: by A7 and binding of Commit^{ivk} its point is $\pm$ the key's $ak^{\mathbb{P}}$, and A6 gives $rk = \pm ak^{\mathbb{P}} + [\alpha] G^{\text{SpendAuth}}$. The signature is then a forgery under re-randomisation, from which forking extracts rsk^* and hence $ask = \pm(rsk^* - \alpha)$: a discrete logarithm of $ak^{\mathbb{P}}$.

Privacy

Theorem. *Under preconditions (honest keys, construction as specified, local paths, no viewing key of a party held), two Ironwood-pool bundles with equal public leakage (pool, number of Actions, anchor, flags, v^{balance}) have computationally indistinguishable public data. In particular a dummy Action is indistinguishable from a real one.*

Proof. a sequence of hybrids, each justified by one assumption:

replace	justified by
the proof by a simulated one	zero knowledge
rk by uniform points	uniform α
nf by $\text{Extract}_{\mathbb{P}}([u] K^{\text{Orchard}} + \text{cm})$	Poseidon PRF, joint hiding
C_{out} by an encryption of a fixed string	PRF^{ock} oracle, AEAD
$\text{epk}^*, C_{\text{enc}}$ by random ones	DDH, KDF oracle, AEAD
cmx by $\text{Extract}_{\mathbb{P}}$ of a uniform point	PRF expansion, then perfect hiding
$(\text{cv}_i^{\text{net}})_i$ and bsk jointly	perfect: bijection fixing $\sum_i \text{rcv}_i$

The end-to-end theorem

For each pool, no efficient adversary achieves any of the following, except with negligible probability:

- (i) an accepted Action with no extractable witness, a second opening of a created cmx, or a consumed note of non-zero value that is not a leaf at the anchor R1, R2
- (ii) two accepted Actions consuming one committed note of non-zero value R3
- (iii) an accepted bundle whose v^{balance} differs from $\sum (v^{\text{old}} - v^{\text{new}})$ R4
- (iv) consuming a note of an honest key without its holder's signature, or changing the signed data of a transaction R5
- (v) for the Ironwood pool, distinguishing two bundles with equal public leakage hiding of R1–R4

Clauses (i)–(iii) use no property of key generation; (iv) and (v) are claimed for keys with `use_qsk = false`.

How Halo 2 proves

PLONKish constraint systems

Over a prime field $\mathbb{F}$, with $n = 2^k$ rows labelled by $H := \{\omega^0, \dots, \omega^{n-1}\}$, $Z_H(X) := X^n - 1$:

1. columns of n cells: *fixed* (the circuit), *advice* (the witness), *instance* (public inputs);
2. custom gates $g_t = 0$ of degree $\leq d_{\max}$ in cells of the current row and of rows at fixed rotations $v(\omega^r X)$;
3. equality constraints: a permutation σ of a set of cells whose cycles must hold equal values;
4. lookups: input expressions $(A_0, \dots, A_{m-1})$ whose every row is a row of a table $(S_0, \dots, S_{m-1})$.

A column is its interpolant: $v(X)$ of degree $< n$ with $v(\omega^i) =$ the cell of row i . A constraint holds on every row iff its polynomial is divisible by Z_H .

The toy: a table

“I know x with $x^3 + x + 5 = 35$ ” over $\mathbb{F}_{97}$; the witness $x = 3$.

$$q_M a b + q_L a + q_R b + q_O c + q_C + \text{PI} = 0 \quad \text{on every row}$$

	advice			fixed					instance	
row	a	b	c	q_M	q_L	q_R	q_O	q_C	PI	
0	3	3	9	1	0	0	-1	0	0	$x \cdot x = x^2$
1	9	3	27	1	0	0	-1	0	0	$x^2 \cdot x = x^3$
2	27	3	30	0	1	1	-1	0	0	$x^3 + x = s$
3	30	0	0	0	1	0	0	5	-35	$s + 5 = 35$

Copy constraints: $a_0 = b_0 = b_1 = b_2 = x$; $c_i = a_{i+1}$.

Rows become polynomials

The element $\omega = 22$ has order 4 in $\mathbb{F}_{97}$: $H = \{1, 22, 96, 75\}$, $Z_H(X) = X^4 - 1$.

$$a(1) = 3, a(22) = 9, a(96) = 27, a(75) = 30 \implies a(X) = 90 + 61X + 22X^2 + 24X^3$$

$$G(X) := q_M ab + q_L a + q_R b + q_O c + q_C + \text{PI}, \quad \deg G \leq 9$$

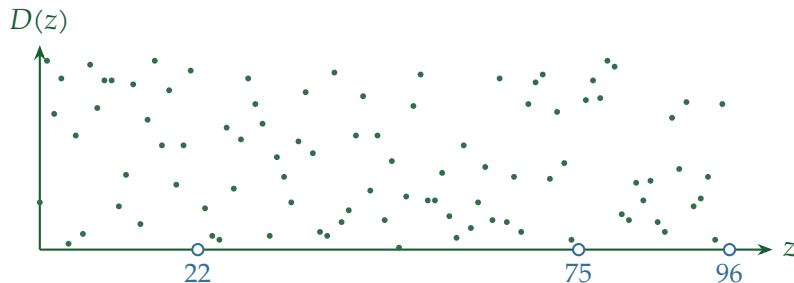
Theorem. *Every gate holds $\iff G$ vanishes on $H \iff Z_H \mid G \iff G = t \cdot Z_H$ for some polynomial t .*

The lie $x^2 = 10$ in row 0 leaves, for any t^* , a nonzero gap $D := G^* - t^* Z_H$; for the Euclidean quotient $D = 24(1 + X + X^2 + X^3)$.

One random point

Theorem (Schwartz–Zippel). *If the trace is invalid and the committed G^*, t^* are fixed before z is drawn uniformly from $\mathbb{F}$, then*

$$\Pr_z[G^*(z) = t^*(z) Z_H(z)] = \Pr_z[D(z) = 0] \leq \frac{\deg D}{|\mathbb{F}|}.$$



Toy: 3/97. Deployed: $\deg D < 9 \cdot 2^{11}$, so $\leq 18432/p_{\text{Pallas}} < 2^{-239}$.

The permutation argument

Cells (i, j) of the m equality columns get labels $\delta^i \omega^j$; fixed polynomials s_i hold the labels permuted by σ . After challenges β, γ the prover commits to Z_{perm} with

$$\begin{aligned} \ell_0(X) (1 - Z_{\text{perm}}(X)) &\equiv 0 \\ Z_{\text{perm}}(\omega X) \prod_i (v_i + \beta s_i + \gamma) - Z_{\text{perm}}(X) \prod_i (v_i + \beta \delta^i X + \gamma) &\equiv 0 \end{aligned}$$

modulo Z_H : together

$$\prod_{i,j} (v_i(\omega^j) + \beta \delta^i \omega^j + \gamma) = \prod_{i,j} (v_i(\omega^j) + \beta s_i(\omega^j) + \gamma).$$

Theorem. *The copy constraints hold iff the two products agree as polynomials in (β, γ) ; a violation survives random β, γ with probability $\leq 2N/|\mathbb{F}|$, N the number of cells.*

The lookup argument

Goal: every value of A on the active rows occurs in S . The prover commits to A' , a permutation of A with equal values in runs, and S' , a permutation of S with each run's value at its first row; after β, γ , to Z_{lookup} :

$$\ell_0(X) (1 - Z_{\text{lookup}}(X)) \equiv 0$$

$$Z_{\text{lookup}}(\omega X) (A' + \beta)(S' + \gamma) - Z_{\text{lookup}}(X) (A + \beta)(S + \gamma) \equiv 0$$

$$\ell_0(X) (A' - S') \equiv 0$$

$$(A' - S') (A'(X) - A'(\omega^{-1}X)) \equiv 0$$

modulo Z_H on the active rows.

Theorem. *If the identities hold, then except with probability $\leq 4n/|\mathbb{F}|$ every value of A on the active rows is a value of S .*

Toy: $S = (0, 1, 2, 3)$, $A = (2, 3, 2, 2)$, $A' = (2, 2, 2, 3)$, $S' = (2, 0, 1, 3)$.

The vanishing argument

All constraints $g_0, \dots, g_{m-1}$ (gates, permutation rules, lookup rules) are combined with a challenge y :

$$h(X) := \frac{\sum_i y^i g_i(X)}{Z_H(X)}, \quad h(X) = \sum_{j=0}^{d_{\max}-2} X^{nj} h_j(X)$$

each h_j of degree $< n$ is committed separately.

Theorem. *The quotient satisfies $\deg h \leq d_{\max}(n-1) - n = (d_{\max} - 1)(n-1) - 1$.*

At a challenge x the prover sends the value at x and at the needed rotations $\omega^r x$ of every committed polynomial except the pieces h_j . The verifier computes $\sum_i y^i g_i(x) / Z_H(x)$, and the opening argument checks it against $\sum_j [x^{nj}] \text{Comm}(h_j)$.

Action circuit: $d_{\max} = 9$, $n = 2^{11}$, so $\deg h \leq 16375$.

The transcript, and Fiat-Shamir

prover sends	then the challenge
commitments to the advice columns	θ
permuted lookup columns A', S'	β, γ
permutation and lookup products Z ; a random polynomial	y
the pieces h_j of the quotient	x
all evaluations at x and $\omega^r x$	x_1, x_2
the multipoint quotient commitment	x_3
the values $q_i(x_3)$	x_4
the mask commitment S	ξ, ζ
(L_j, R_j) , one pair per round	u_j

Each challenge is a hash of the transcript so far: the verifier's coins become a function of everything the prover has committed to.

Many openings to one

Polynomials p_ν with commitments C_ν , grouped by their query sets $T^{(i)}$,
 $Z_i(X) := \prod_{t \in T^{(i)}} (X - t)$:

1. Challenge x_1 : $q_i := \sum_e x_1^e p_{\nu_{i,e}}$, $Q_i := \sum_e [x_1^e] C_{\nu_{i,e}}$; r_i interpolates the claimed values on $T^{(i)}$;
2. Challenge x_2 : $f := \sum_i x_2^{i-1} (q_i - r_i) / Z_i$, committed as F ;
3. Challenge x_3 : the prover sends $u_i := q_i(x_3)$; the verifier forms
$$v_f := \sum_i x_2^{i-1} \frac{u_i - r_i(x_3)}{Z_i(x_3)};$$
4. Challenge x_4 : $f^* := f + \sum_i x_4^i q_i$, $F^* := F + \sum_i [x_4^i] Q_i$, $v^* := v_f + \sum_i x_4^i u_i$.

One opening of F^* at x_3 to v^* replaces them all: f is a polynomial only if every claimed value is true.

The inner-product commitment

Generators $\mathbf{G} = (G_0, \dots, G_{d-1}), H, U$ on Vesta, from a hash; $d = 2^k$.

$$\text{Comm}(p; r) := \langle \mathbf{a}, \mathbf{G} \rangle + [r] H, \quad p(X) = \sum_{i < d} a_i X^i$$

$$\mathcal{R}_{\text{IPA}} := \{((P, x, v); (\mathbf{a}, r)) : P = \langle \mathbf{a}, \mathbf{G} \rangle + [r] H, v = \langle \mathbf{a}, \mathbf{b}(x) \rangle\}$$

$$\mathbf{b}(x) := (1, x, \dots, x^{d-1})$$

$a_i, r, x, v \in \mathbb{F}_{p_{\text{Pallas}}}$, the scalar field of Vesta; $P, \mathbf{G}, H, U \in \mathcal{E}(\mathbb{F}_{p_{\text{Vesta}}})$.

Pedersen on the coefficient vector: perfectly hiding, binding under discrete logarithms on Vesta. Opening is an inner product with a public vector. The problem: prove it with fewer than d group elements.

One fold

Split every vector into halves. The prover sends, with fresh blinders ℓ, ρ ,

$$L := \langle \mathbf{a}_{\text{lo}}, \mathbf{G}_{\text{hi}} \rangle + [\ell] H + [\langle \mathbf{a}_{\text{lo}}, \mathbf{b}_{\text{hi}} \rangle] U$$

$$R := \langle \mathbf{a}_{\text{hi}}, \mathbf{G}_{\text{lo}} \rangle + [\rho] H + [\langle \mathbf{a}_{\text{hi}}, \mathbf{b}_{\text{lo}} \rangle] U$$

Challenge $u \in \mathbb{F}^\times$; both sides fold:

$$\mathbf{a}' := u \mathbf{a}_{\text{lo}} + u^{-1} \mathbf{a}_{\text{hi}}, \quad \mathbf{G}' := u^{-1} \mathbf{G}_{\text{lo}} + u \mathbf{G}_{\text{hi}}, \quad \mathbf{b}' := u^{-1} \mathbf{b}_{\text{lo}} + u \mathbf{b}_{\text{hi}}$$

$$P + [v] U + [u^2] L + [u^{-2}] R = \langle \mathbf{a}', \mathbf{G}' \rangle + [r + u^2 \ell + u^{-2} \rho] H + [\langle \mathbf{a}', \mathbf{b}' \rangle] U$$

The same claim at half the length; the verifier folds $\mathbf{G}$ and $\mathbf{b}$ itself and never sees $\mathbf{a}$.

The recursion and the final check

After rounds $j = k, \dots, 1$ the vectors have length one; the prover sends a and $r^* := r + \sum_j (u_j^2 \ell_j + u_j^{-2} \rho_j)$. Accept iff

$$P + [v] U + \sum_{j=1}^k ([u_j^2] L_j + [u_j^{-2}] R_j) = [a] G^{(0)} + [r^*] H + [a b^{(0)}] U.$$

Theorem (structured scalars). With $i = \sum_{\ell} \beta_{\ell} 2^{\ell}$ and $s_i := \prod_{\ell=1}^k u_{\ell}^{\pm 1}$ (+ if $\beta_{\ell-1} = 1$),

$$G^{(0)} = \langle \mathbf{s}, \mathbf{G} \rangle, \quad b^{(0)} = \langle \mathbf{s}, \mathbf{b}(x) \rangle = \prod_{\ell=1}^k (u_{\ell}^{-1} + u_{\ell} x^{2^{\ell-1}}).$$

The scalar $b^{(0)}$ costs $O(k)$; the point $G^{(0)}$ is one multi-scalar multiplication of size d , the verifier's only linear cost. Proof: $2k$ points, 2 scalars.

Why the argument is sound

Lemma (one round). *From three accepting continuations of one round, with challenges whose squares are distinct, a representation of L , C and R over $(\mathbf{G}, H, U)$ is obtained by solving the 3×3 system with rows $(v^2, 1, v^{-2})$.*

Theorem (knowledge soundness). *There is an extractor that, rewinding an accepting prover along a tree of 3^k transcripts, outputs a witness $(\mathbf{a}, r)$ of $\mathcal{R}_{\text{IPA}}$ or a non-trivial discrete-logarithm relation among $\mathbf{G}, H, U$. The knowledge error is $O(\log d / |\mathbb{F}|)$.*

After Fiat–Shamir the naive reduction loses a factor Q^μ : a factor Q , the number of hash queries, per challenge round, super-polynomial overall. Tight state-restoration bounds are proved in the algebraic group model for the abstract protocol, not for the deployed transcript.

Zero knowledge

Lemma (blinding rows). *Let a column hold uniform independent values on b unused rows. Its evaluations at $m \leq b$ distinct points outside H are jointly uniform and independent of the other rows.*

- Evaluations at x and $\omega^r x$ leak nothing: each advice column has more blinding rows than it is opened at.
- Commitments carry a uniform $[r] H$: perfectly hiding.
- The inner-product rounds carry fresh ℓ_j, ρ_j ; the final message is masked by a random polynomial committed in advance.

Result: a simulator that knows only the instance and programs the hash.

The Action circuit

- The circuit has $n = 2^{11}$ rows over $\mathbb{F}_{p_{\text{Pallas}}}$, so Pallas points are native values; commitments on Vesta; maximum constraint degree 9.
- Sinsemilla by a 2^{10} -row generator table; ten-bit range checks by a lookup; Poseidon rounds as custom gates; incomplete and complete addition and fixed-base multiplication as gates.
- The instance: rt , cv^{net} , nf^{old} , rk , cmx and the three flag inputs, per Action.
- Generators from GroupHash on Vesta: no trusted setup.
- Proof size $2720 + 2272a$ bytes for a Actions: 4992 for one, 7264 for two; the inner-product share is 23 points and 2 scalars, 800 bytes.

The whole picture

check	where	mathematics
the note exists	proof: A1, A3	Sinsemilla binding, Merkle fold
spent once	proof A5; nullifier set	keyed PRF, $\text{Extract}_{\mathbb{P}}([s] K^{\text{Orchard}} + \text{cm})$
value conserved	proof A4; binding sig.	$\sum \text{cv}^{\text{net}} - [v^{\text{balance}}] V^{\text{Orchard}} = [\text{bsk}] R^{\text{Orchard}}$
authorised	proof A6, A7; spend sig.	$\text{rk} = \text{ak}^{\mathbb{P}} + [\alpha] G^{\text{SpendAuth}}$
hidden	everything above	hiding, PRFs, DDH, zero knowledge

A verifier learns that every check passes, the balancing value, the anchor and the flags, and nothing about the notes.